
CONTEXTUAL LEARNING FOR INTERRUPTION TOLERANCE AND MULTITASKING

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ABSTRACT

This project is an effort to combine the Morphognosis and Mona neural network models into a comprehensive model for learning and behavior called Mandala. Mona features a contextual causation learning with goal-directed motivation. Morphognosis features contextual multilayer perceptron (MLP) learning. Mandala achieves this by accumulating contextual tiers of temporal states that are fed into an MLP at each time step. Natural environments abound in event streams that require interruption tolerance and multitasking. Mandala affords multitasking as it is robust in the presence of intervening events from overlaid causation chains, a capability that conventional recurrent artificial neural networks (RNNs) struggle with. The results bear this out. In addition, accumulating temporal information discretely labels hierarchical cause-and-effect relationships that can be used for augmented processing. In the case of Mona, channeling motivation through the network for the purpose of goal-seeking requires this feature.

Keywords: Mandala, causation learning, multitasking, artificial neural network, machine learning.

INTRODUCTION

A mated pair of eagles, Jackie and Shadow, are popular on the internet these days as viewed with a hidden camera and microphone (Forgione, 2020). They spend their days and nights building and repairing their nest, brooding eggs, hunting, and fending off threats. In one segment Shadow is seen screaming for Jackie to help with intruding ravens that have in the past destroyed their eggs. He knows he must remain sitting atop them. She has been gone overlong and he must be hungry. This in general is a typical condition for animals and the neural networks that orchestrate them. Multitasking in the midst of interruptions and interferences between goals.

Artificial neural networks (ANNs) currently deal with what might be termed controlled surprise. Self-driving cars are in the focused business of driving. They can handle unexpected road obstructions which they have previously seen in training. Large Language Models (LLMs) (Yin, et al., 2024) can sensibly respond to just about any query, but while composing said response they do not deal with a crying baby or a flat tire. Human animals do.

This project is an effort to find a way how interruptions can be accommodated and multitasking can be accomplished more effectively by an ANN. The resulting architecture is named Mandala, after a geometric pattern that is typically composed of concentric rings or tiers of symbols having some comprehensive meaning (Merriam–Webster Online Dictionary, 2008). Here Mandala is also a tiered construction, where higher tiers represent longer-term context that aids in prediction.

Mandala is also a step toward synthesizing the Morphognosis (Portegys, 2022) and Mona (Portegys, 2010 and 2024b) neural network models into a comprehensive model for learning and behavior. Mona features a contextual causation learning with goal-directed motivation. Morphognosis features contextual multilayer perceptron (MLP) learning. Mandala achieves this by accumulating contextual tiers of temporal states that are fed into an MLP at each time step. In addition, accumulating non-hidden temporal information discretely labels hierarchical cause-and-effect relationships that can be used for augmented processing. In the case of Mona, channeling motivation through the network for the purpose of goal-seeking requires this feature.

Architecturally Mandala bears a resemblance to a Jordan artificial neural network (Jordan, 1986), as shown in Figure 1.

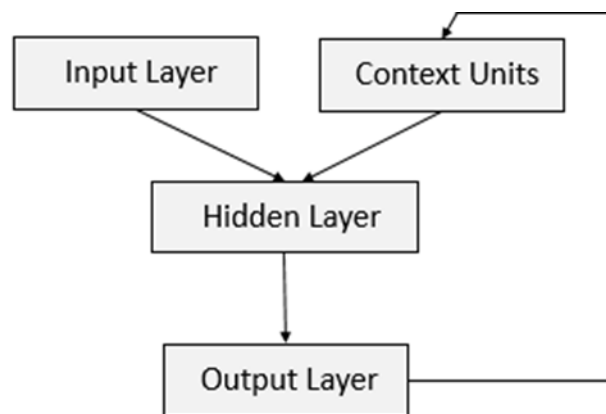


Figure 1 – Jordan network

In a Jordan network, the output layer is fed into context units, which in turn are fed into the hidden layer along with the input layer. In Mandala, the output layer produces not only a prediction, but also temporal state, which after processing combines with external input.

Mandala also shares some commonalities with networks that make use functionality external to the core network. Some of these are discussed below.

Recurrent Neural Networks (RNNs) with external temporal memory are architectures designed to overcome the limitations of standard RNNs (like LSTMs or GRUs) in memorizing long-term dependencies. While standard RNNs rely on internal hidden states that often suffer from

vanishing gradients over long sequences, these models integrate a separate, addressable memory bank to store and retrieve past information over extended time horizons.

Neural Turing Machines (NTMs) (Graves et al., 2014). This is a model that couples a neural network controller to an external memory bank, capable of learning to manipulate and manage the memory to solve tasks like sorting or copying.

A Dynamic Memory Network (DMN) (Kumar et al., 2016) is a neural network architecture that uses an external episodic memory to answer questions about input sequences (stories, text).

External Addressable Long-Term and Working Memory (EALWM) (Quan et al., 2020) splits external memory into distinct long-term and working memory parts, improving the ability to learn long-term dependencies without gradient issues.

The above models retain an RNN at their core, using external storage as an adjunct memory. In Mandala external memory serves as the sole repository of temporal state, combining environment and state information as input to a non-recurrent MLP.

DESCRIPTION

PROBLEM GENERATION

Training and testing event streams are generated using randomly generated grammars that produce binary hierarchies of cause-and-effect pairs of terminals. A simple such hierarchy is presented in Figure 2.

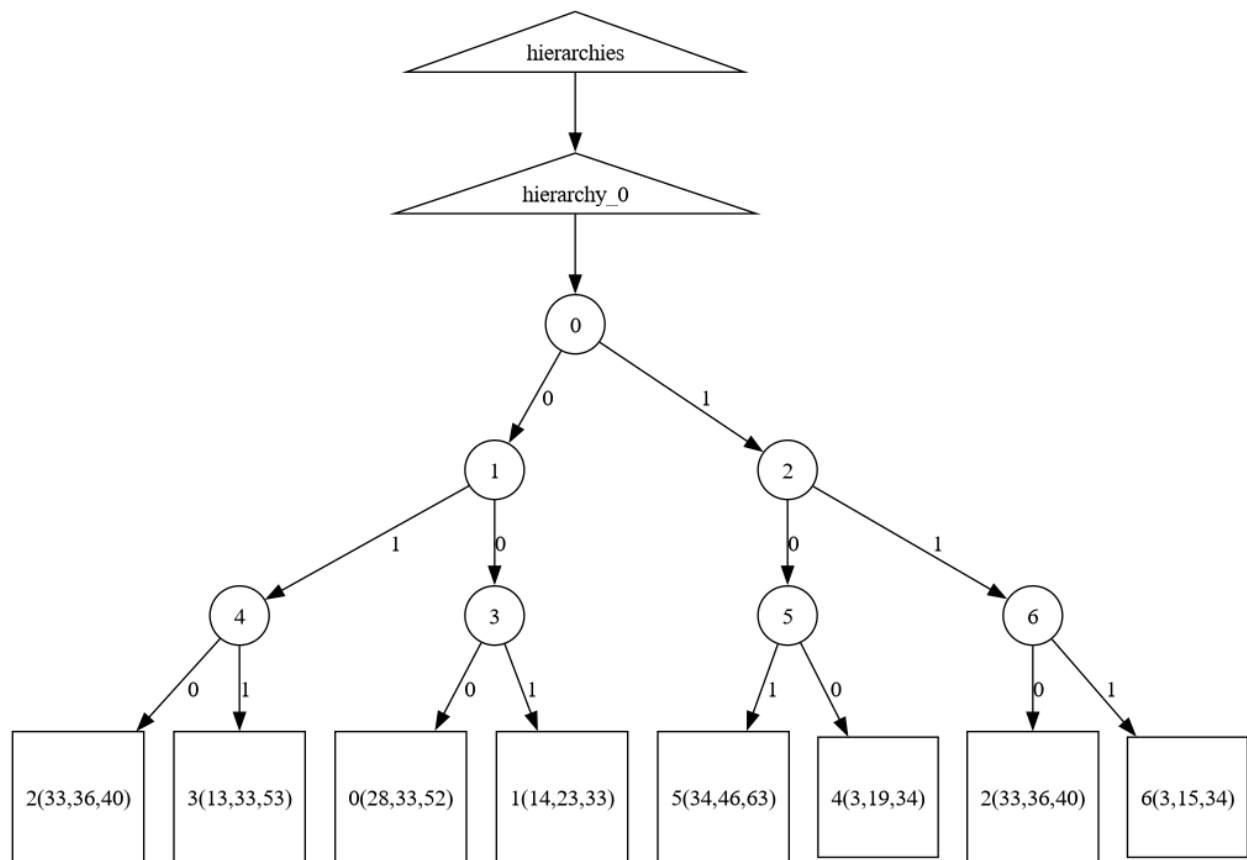


Figure 2 – Simple generated hierarchy

Here there are seven non-terminals (0-6), indicated by circles, and seven terminals, indicated by squares. The numbers on the edges indicate the child order. The parenthesized numbers in the terminals are sparse encodings for the neural network, to be discussed later.

This hierarchy produces a path of: {0, 1, 2, 3, 4, 5, 2, 6}.

The terminal children of a non-terminal are considered to be a cause-and-effect pair. The goal of the neural network is to predict the effect given the cause as they are presented in the input stream.

Generating inputs using this method presents context-related challenges. For example, the appearance of cause 2 cannot solely be used to determine which of the effects 3 or 6 is correct. The correct prediction depends on the context provided by the previous cause-and-effect pair.

Parameters determine the shape of a hierarchy:

- NUM_CAUSATION_HIERARCHIES
- NUM_NONTERMINALS

- NUM_TERMINALS
- TERMINAL_PRODUCTION_PROBABILITY

Multiple paths are generated by multiple hierarchies. Node ids are scoped by hierarchy, so the same id in different hierarchies will be unique to each hierarchy. The terminal production probability affects the hierarchy height: a greater probability causes non-terminals to be less likely to produce non-terminal children, thus growing the hierarchy further.

An example of two hierarchies having identical structure is presented in Figure 3.

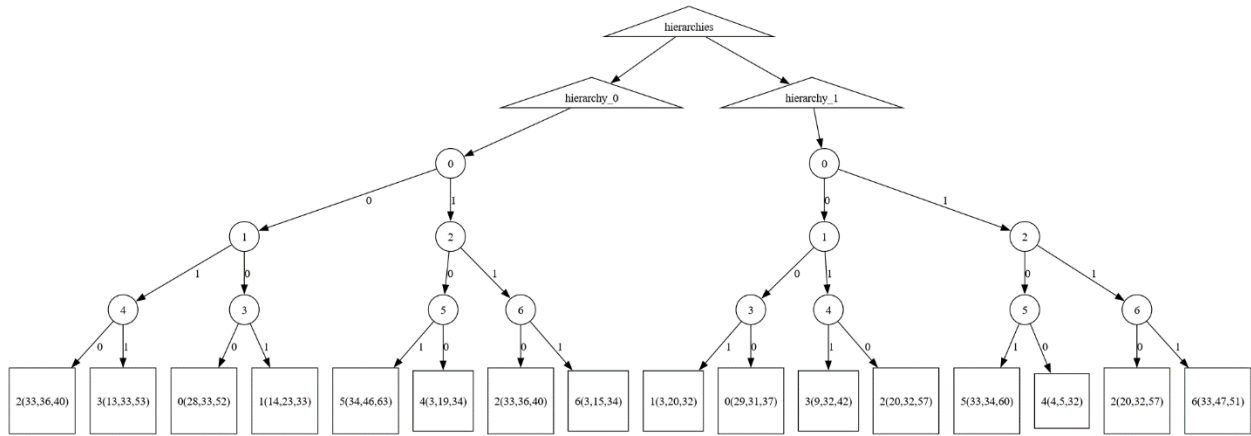


Figure 3 – Two hierarchies

Note that a cause terminal is not caused, which means that temporal gaps containing interstitial events can precede it, presenting another learning challenge. For example, with interstitial events the stream given by the hierarchy in Figure 1 might look like:

{a, b, 0, 1, b, 2, 3, b, c, 4, 5, 2, 6}

Where the letters a-c represent interstitial events.

A final challenge, addressing multitasking, can be shown by interleaving the streams from the hierarchies in Figure 2:

{0/0, 1/0, 2/0, 3/0, 0/1, 1/1, 4/0, 5/0, 2/1, 3/1, 2/0, 6/0, 4/1, 5/1, 2/1, 6/1}

Where the hierarchy number is appended to the terminal id after a slash ("/") character.

ALGORITHM

The generated terminal ids are encoded into a sparse representation of one-hot inputs to the neural network. Sparse encoding is also used to represent higher tiered information. The advantage of sparsity in neural networks is delineated in the Numenta White Paper (2021).

During training, paths are presented to the network without interstitial or interleaved events. Those are supplied during testing. This is how training is done:

The input to the network at time t is denoted X_t . The output is y_t . Concatenated to the external input and prediction output, there are n tiers of encoded temporal states.

The tier values for y_t are computed as follows:

```
update_y_tiers:
    tier_A = encode(external input)
    for tier i from 0 to n:
        tier_B =  $X_t$  values of tier i
        if tier_B exists:
            negate values for tier_B in tier i
            tier_A = encode(tier_A, tier_B)
        else:
            enter values for tier_A in tier i
            break
```

The tier values of $X_{t+1} = X_t + y_t$

The network is thus trained to compute temporal context information.

Here is an example scenario:

```
t = 0, X: [sense=[4, 14, 41]], y: [sense=[21, 24, 40], tier0=[4, 14, 41]]

t = 1, X: [sense=[21, 24, 40], tier0=[4, 14, 41]], y: [sense=[5, 42, 43], tier0=[-4, -14, -41], tier1=[8, 9, 34]]

t = 2, X: [sense=[5, 42, 43], tier0=[], tier1=[8, 9, 34]], y: [sense=[38, 40, 42], tier0=[5, 42, 43]]
```

Values time-out in proportion to their tier: values in higher tiers persist longer.

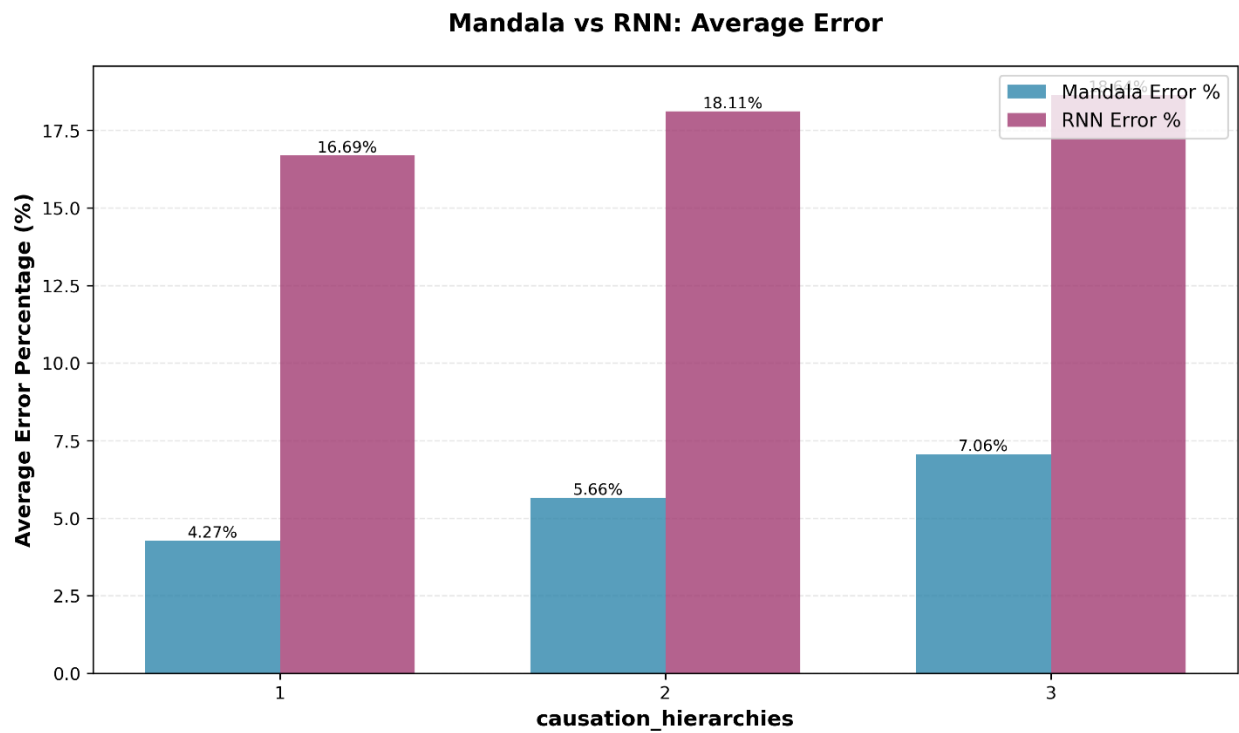
For testing, interstitial and interleaved events are also presented to the network.

Note that the algorithm could also be used to recognize out-of-order streams, a topic explored by Portegys (2024a).

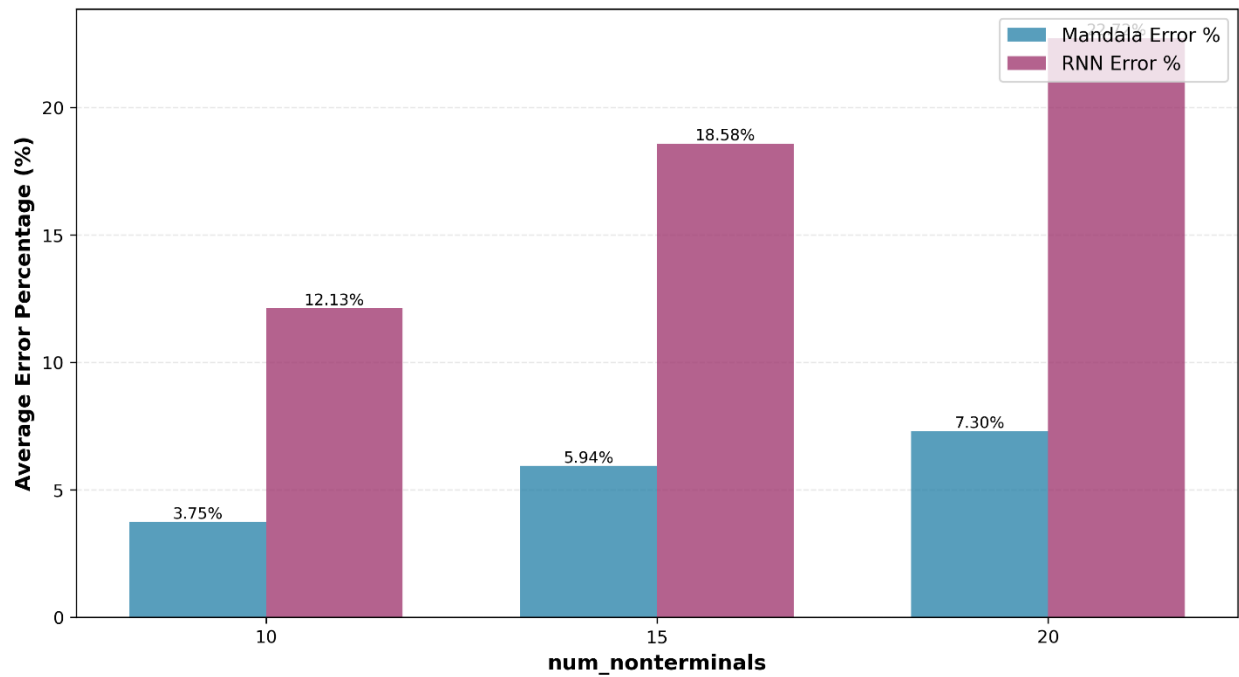
The code is available on GitHub at: <https://github.com/morphognosis/Mandala>

RESULTS

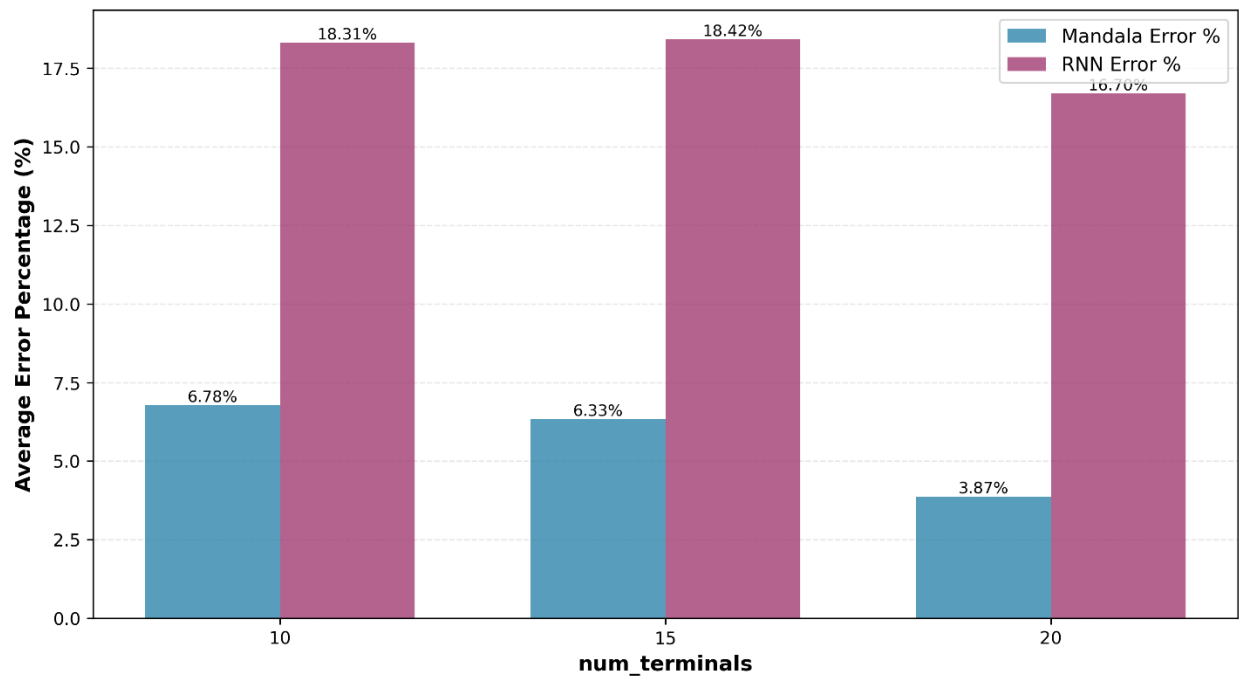
MANDALA/RNN COMPARISONS



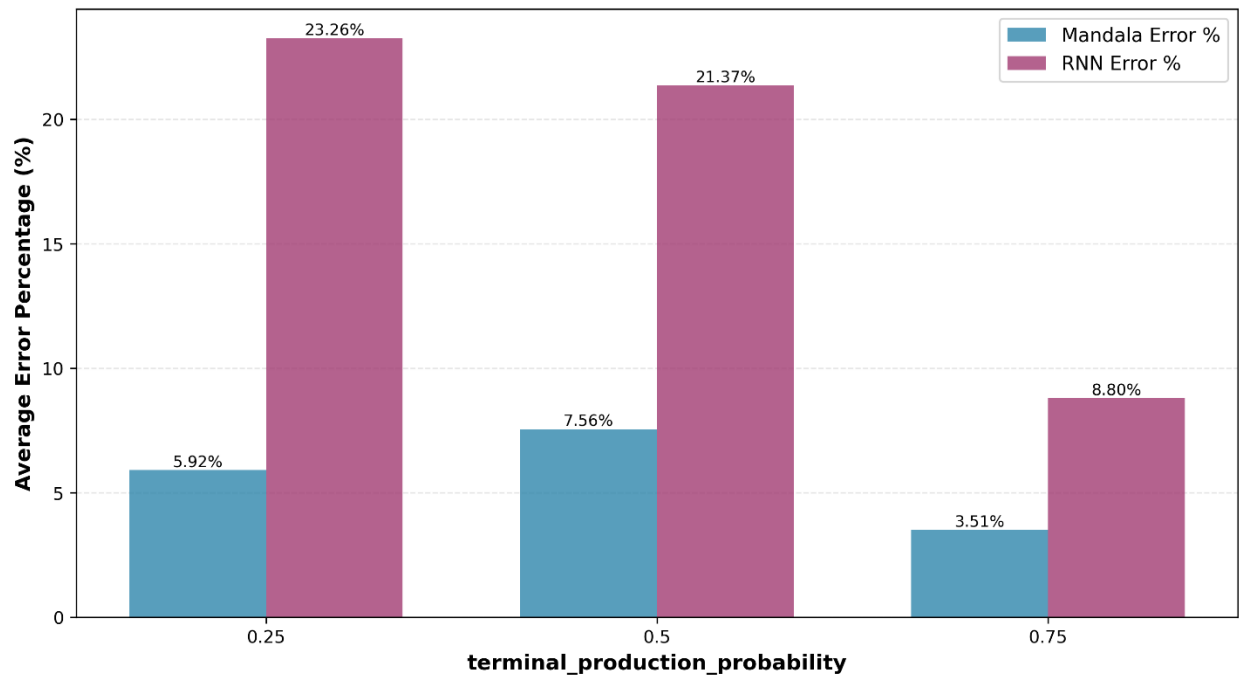
Mandala vs RNN: Average Error



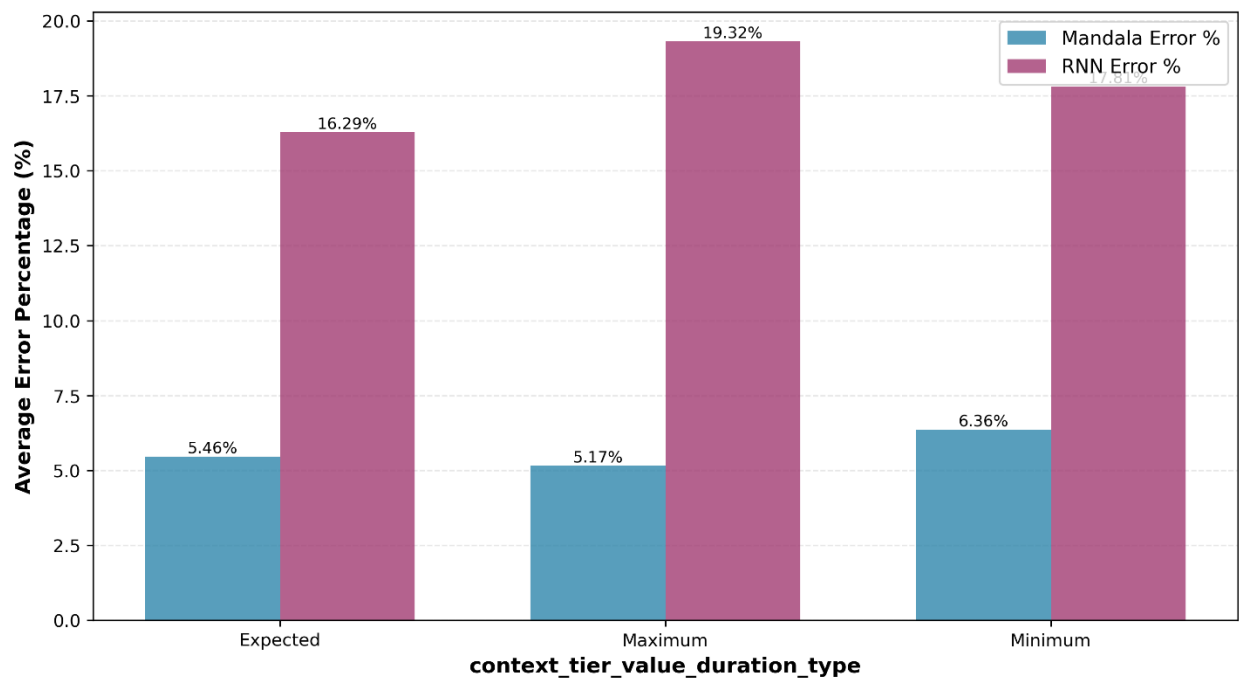
Mandala vs RNN: Average Error

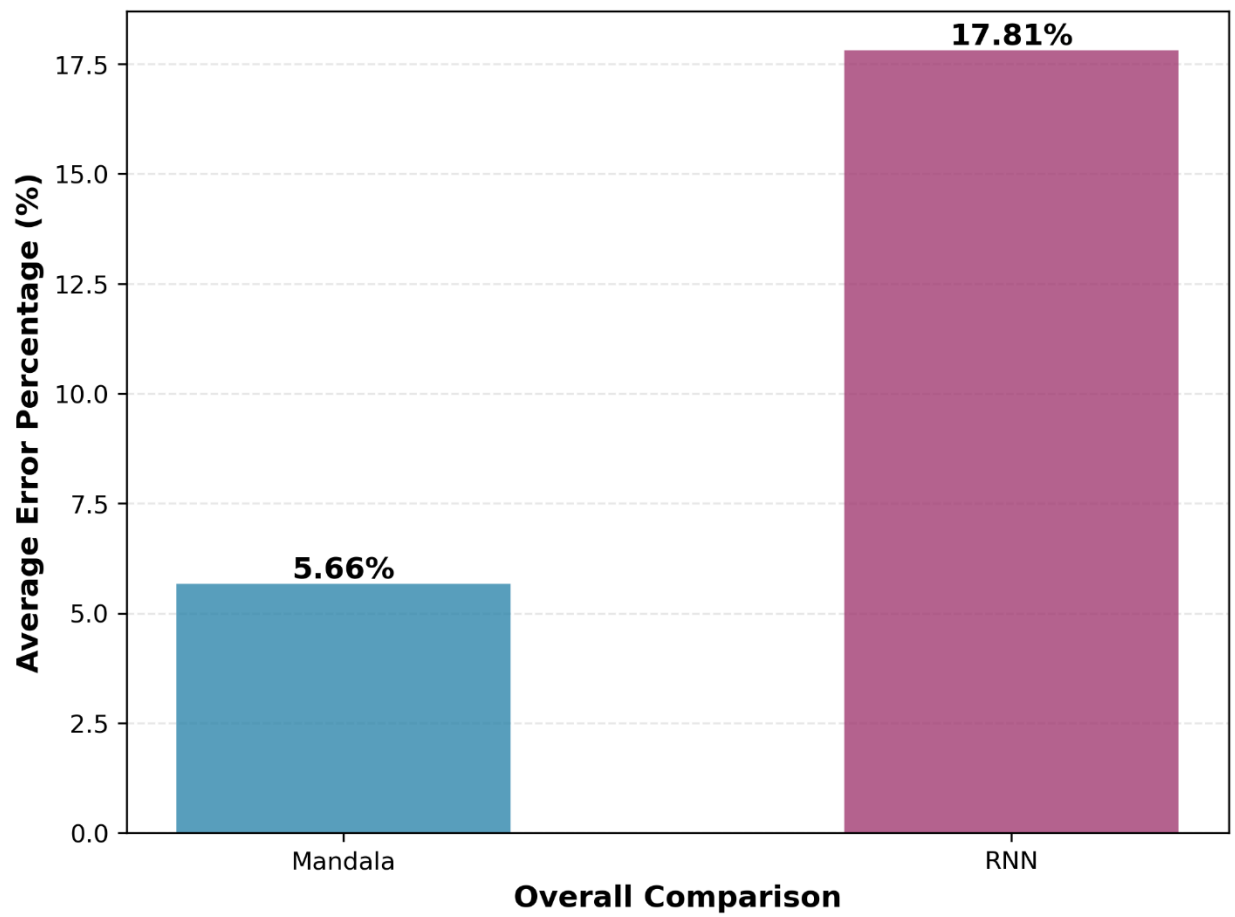


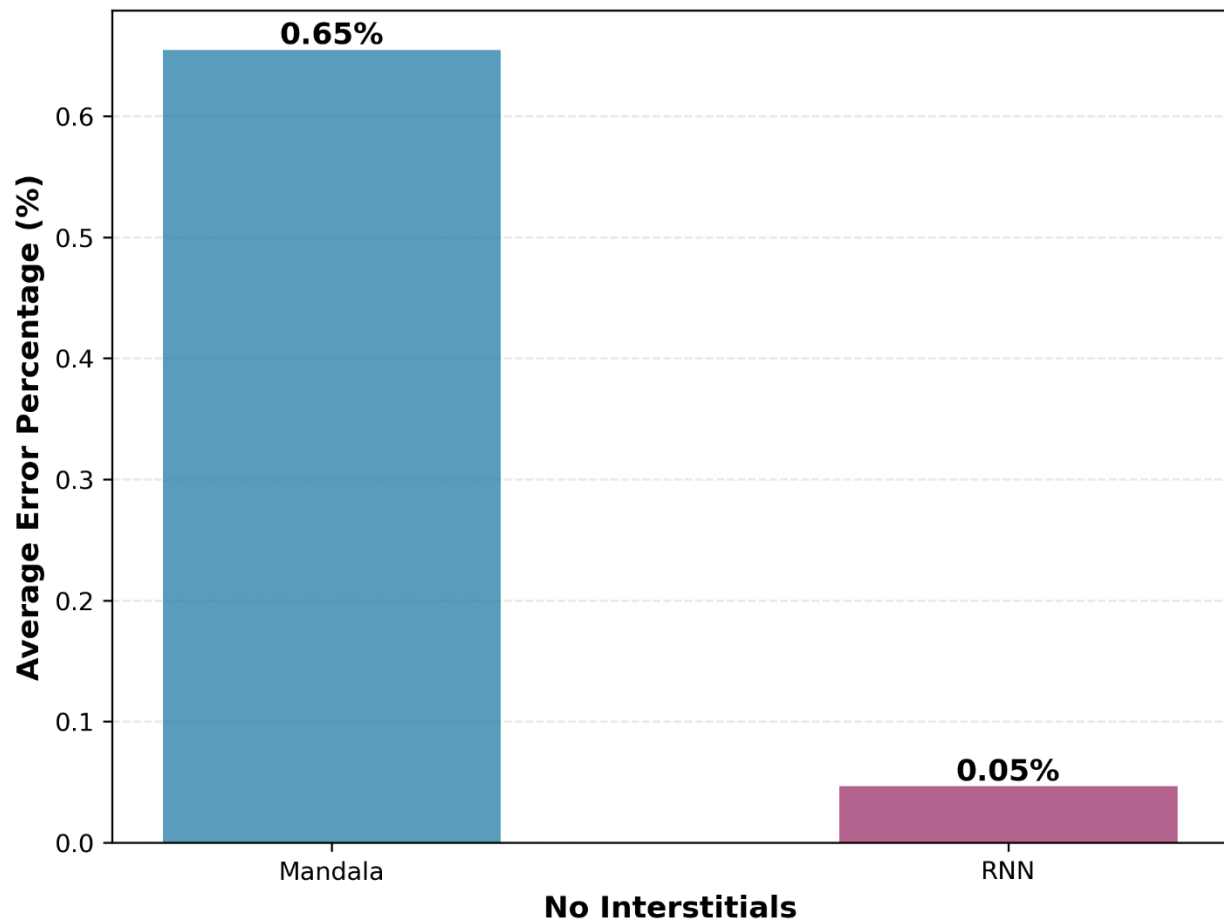
Mandala vs RNN: Average Error



Mandala vs RNN: Average Error







LLM TESTING

Tried Claude, Gemini, ChatGPT on test_0 sequences.

Given these sequences:

0,1,2,3

4,5,2,6

Predict the next number in these sequences:

8,8,7,0

7,11,4

7,4,5,11,11,8,10,10,10,11,2

0,1,9,11,10,2

7,9,11,9,7,8,4,5,7,10,10,8,7,2

7,0,1,11,10,7,7,10,2

4,5,10,8,8,2

11,8,0,1,8,10,2

11,7,7,11,11,0,1,11,7,11,11,9,11,10,7,2

10,8,9,7,4,5,9,8,8,10,11,11,7,9,9,11,2

Note that 4,5 causes 2,3 and 0,1 causes 2,6

Claude predicted the first two simple sequences, But failed on the sequence that depended on recognizing the earlier transitions determine the ending transitions.

ChatGPT did not correctly predict any of the sequences.

CONCLUSION

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